On Hypergeometric Series of Eisenstein Type

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Outline

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An index $\boldsymbol{k} = (k_1, \ldots, k_r)$ is *admissible* if $k_r \geq 2$. The index of depth zero is called the *empty index* and is denoted by $\varnothing$.

Definition (Multiple zeta values)

For an admissible index $\boldsymbol{k} = (k_1, \ldots, k_r)$, define

$$\zeta(\boldsymbol{k}) = \sum_{0<n_1<\cdots<n_r} \frac{1}{n_1^{k_1} \cdots n_r^{k_r}}.$$

For the empty index, set $\zeta(\varnothing) = 1$.

For $\boldsymbol{k} = (k_1, \ldots, k_r)$, its *weight* is $k_1 + \cdots + k_r$, its *depth* is r, and its *height* is the number of components $k_i \geq 2$.

Rising factorials

Let $I(k,r,s)$ and $I_0(k,r,s)$ denote the sets of all indices and all admissible indices of weight k, depth r, and height s, respectively.

Definition (Rising factorials)

For an integer n and a complex number a, define

$$(a)_n := \begin{cases} a(a+1)\cdots(a+n-1), & n>0, \\ 1, & n=0, \\ \dfrac{(-1)^n}{(1-a)\cdots(-n-a)}, & n<0. \end{cases}$$

We also write $(a_1,\dots,a_r)_n = (a_1)_n \cdots (a_r)_n$.

q-shifted factorials

Let $|q| < 1$.

Definition (q-shifted factorials)

For a nonnegative integer n and a complex number a, define

$$(a;q)_n := \begin{cases} (1-a)(1-aq)\cdots(1-aq^{n-1}), & n > 0, \\ 1, & n = 0. \end{cases}$$

Set

$$(a;q)_\infty := \lim_{n\to\infty}(a;q)_n = \prod_{k\geq 0}(1-aq^k).$$

We also write $(a_1,\ldots,a_r;q)_n = (a_1;q)_n\cdots(a_r;q)_n$.

Definition (Hypergeometric and basic hypergeometric series)

Define the hypergeometric and basic hypergeometric series by

$$ {}_2F_1\left[\begin{matrix} a, b \\ c \end{matrix}; x\right] := \sum_{n\geq 0} \frac{(a,b)_n}{n!(c)_n} x^n, $$

$$ {}_2\phi_1\left[\begin{matrix} a, b \\ c \end{matrix}; q, x\right] := \sum_{n\geq 0} \frac{(a,b;q)_n}{(c,q;q)_n} x^n. $$

The Ohno–Zagier relation

The following theorem of Ohno and Zagier relates multiple zeta values to hypergeometric series.

Theorem (Ohno–Zagier relation)

Let $\alpha+\beta=u+v$ and $\alpha\beta=w$. As a formal power series in u, v, w,

$$\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)}\zeta(\boldsymbol{k})\right)u^{k-r-s}v^{r-s}w^{s-1}=\frac{1}{uv-w}\left(1-{}_2F_1\left[\begin{matrix}\alpha-u,\beta-u\\1-u\end{matrix};1\right]\right)$$

$$=\frac{1}{uv-w}\left(1-\frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-\alpha)\Gamma(1-\beta)}\right).$$

The Ohno–Zagier relation

The gamma function has the expansion

$$\Gamma(1-x) = \exp\left(\gamma x + \sum_{n\geq 2} \frac{\zeta(n)}{n} x^n\right).$$

Thus the Ohno–Zagier relation shows that the sum of multiple zeta values of fixed weight, depth, and height,

$$\sum_{\boldsymbol{k}\in I_0(k,r,s)} \zeta(\boldsymbol{k}),$$

can be expressed in terms of Riemann zeta values.

Definition (Multiple Eisenstein series)

Let $\Im\tau > 0$. Define an order $\prec$ on $\mathbb{Z} + \mathbb{Z}\tau$ by

$$m\tau + n \prec m'\tau + n' \quad \Longleftrightarrow \quad m < m' \text{ or } (m = m' \text{ and } n < n').$$

For an index $\boldsymbol{k} = (k_1, \ldots, k_r)$ with all $k_i \geq 2$, define

$$G(\boldsymbol{k}; \tau) := \lim_{N \to \infty} \sum_{\substack{0 \prec \lambda_1 \prec \cdots \prec \lambda_r \prec N\tau \\ \lambda_1, \ldots, \lambda_r \in \mathbb{Z} + \mathbb{Z}\tau}} \frac{1}{\lambda_1^{k_1} \cdots \lambda_r^{k_r}}.$$

Set $G(\varnothing; \tau) = 1$.

We now introduce a regularization that extends this definition to arbitrary admissible indices.

Multitangent functions

For an index $\boldsymbol{k} = (k_1, \ldots, k_r)$ with $k_1, k_r \geq 2$ and $\Im t > 0$, define the multitangent function by

$$\Psi_{\boldsymbol{k}}(t) := \sum_{\substack{n_1 < \cdots < n_r \\ n_1, \ldots, n_r \in \mathbb{Z}}} \frac{1}{(n_1 + t)^{k_1} \cdots (n_r + t)^{k_r}}.$$

For an index $\boldsymbol{k}$ with all components at least 2, define

$$h(\boldsymbol{k}; \tau) := \sum_{\substack{j \geq 1 \\ (\boldsymbol{k}_1, \ldots, \boldsymbol{k}_j) = \boldsymbol{k} \\ \boldsymbol{k}_1, \ldots, \boldsymbol{k}_j \neq \varnothing}} \sum_{0 < l_1 < \cdots < l_j} \Psi_{\boldsymbol{k}_1}(l_1 \tau) \cdots \Psi_{\boldsymbol{k}_j}(l_j \tau).$$

Set $h(\varnothing; \tau) = 1$.

A representation of G

The following representation is known.

Proposition (Bachmann 2012)

For an index $\boldsymbol{k} = (k_1, \ldots, k_r)$ *with all* $k_i \geq 2$,

$$G(\boldsymbol{k}; \tau) = \sum_{i=0}^{r} \zeta(k_1, \ldots, k_i)\, h(k_{i+1}, \ldots, k_r; \tau).$$

Multiple harmonic sums of Hurwitz type

For an index $\boldsymbol{k} = (k_1, \ldots, k_r)$, define

$$\zeta_N(\boldsymbol{k}; t) := \sum_{0<n_1<\cdots<n_r<N} \frac{1}{(n_1+t)^{k_1}\cdots(n_r+t)^{k_r}},$$

and write $\zeta_N(\boldsymbol{k}) := \zeta_N(\boldsymbol{k}; 0)$. For an admissible index $\boldsymbol{k}$, set

$$\zeta(\boldsymbol{k}; t) := \lim_{N\to\infty} \zeta_N(\boldsymbol{k}; t).$$

Splitting the summation for the multitangent function into the nonpositive and positive parts gives

$$\Psi_{k_1,\ldots,k_r}(t) = \sum_{i=0}^{r} (-1)^{k_1+\cdots+k_i} \zeta(k_i, \ldots, k_1; -t-1)\zeta(k_{i+1}, \ldots, k_r; t).$$

We use the following regularization theorem.

Proposition (Kaneko–Xu–Yamamoto 2021)

For every index $\boldsymbol{k}$, there exist $J > 0$ and a polynomial $P_{\boldsymbol{k};t}(T) \in \mathbb{C}[T]$ such that, as $N \to \infty$,

$$\zeta_N(\boldsymbol{k}; t) = P_{\boldsymbol{k};t}\big(\zeta_N(1)\big) + O\left(\frac{\log^J N}{N}\right).$$

We write

$$\zeta^T(\boldsymbol{k}; t) := P_{\boldsymbol{k};t}(T).$$

For the empty index, set $\zeta^T(\varnothing; t) := 1$. We also write $\zeta^*(\boldsymbol{k}; t) := \zeta^0(\boldsymbol{k}; t)$ and $\zeta^*(\boldsymbol{k}) := \zeta^*(\boldsymbol{k}; 0)$.

Regularization of multitangent functions

For an index $\boldsymbol{k} = (k_1, \ldots, k_r)$, define the regularized multitangent function by

$$\Psi^T_{k_1,\ldots,k_r}(t) := \sum_{i=0}^{r} (-1)^{k_1+\cdots+k_i} \zeta^{-T}(k_i, \ldots, k_1; -t-1)\zeta^*(k_{i+1}, \ldots, k_r; t).$$

This normalization gives $\Psi^T_1(t) = \pi \cot \pi t + T$. For an admissible index $\boldsymbol{k}$, define the regularized version of h by

$$h^T(\boldsymbol{k}; \tau) := \sum_{\substack{j \geq 1 \\ (\boldsymbol{k}_1,\ldots,\boldsymbol{k}_j)=\boldsymbol{k} \\ \boldsymbol{k}_1,\ldots,\boldsymbol{k}_j \neq \varnothing}} \sum_{0<l_1<\cdots<l_j} \Psi^T_{\boldsymbol{k}_1}(l_1\tau) \cdots \Psi^T_{\boldsymbol{k}_j}(l_j\tau).$$

Set $h^T(\varnothing; \tau) = 1$.

Regularization of multiple Eisenstein series

For an admissible index $\boldsymbol{k} = (k_1, \ldots, k_r)$, define the regularized multiple Eisenstein series by

$$G^T(\boldsymbol{k}; \tau) := \sum_{i=0}^{r} \zeta^*(k_1, \ldots, k_i)\, h^T(k_{i+1}, \ldots, k_r; \tau).$$

In particular, when $T = -\pi i$, we write

$$G^-(\boldsymbol{k}; \tau) := G^{-\pi i}(\boldsymbol{k}; \tau).$$

Hypergeometric series of Eisenstein type

We now introduce an Eisenstein-type analogue of the hypergeometric series at 1,

$$ {}_2F_1\left[\begin{matrix} a, b \\ c \end{matrix}; 1\right] = \sum_{n \geq 0} \prod_{j=0}^{n-1} \frac{(a+j)(b+j)}{(1+j)(c+j)}. $$

For $\lambda \in \mathbb{Z} + \mathbb{Z}\tau$, write $0 \preceq \lambda$ if $\lambda = 0$ or $0 \prec \lambda$. A naive analogue would be

$$ \sum_{0 \preceq \lambda} \prod_{0 \preceq \mu \prec \lambda} \frac{(a+\mu)(b+\mu)}{(1+\mu)(c+\mu)}. $$

Since the product generally fails to converge, we regularize it.

A formal calculation

Writing $\lambda = m\tau + n$, we formally obtain

$$\prod_{0 \preceq \mu \prec \lambda} \frac{a+\mu}{b+\mu} = \left(\prod_{k \geq 0} \frac{a+k}{b+k} \right) \left(\prod_{j=1}^{m-1} \prod_{k \in \mathbb{Z}} \frac{a+j\tau+k}{b+j\tau+k} \right) \left(\prod_{\substack{j<n \\ j \in \mathbb{Z}}} \frac{a+m\tau+j}{b+m\tau+j} \right)$$

$$= \left(\prod_{k \geq 0} \frac{a+k}{b+k} \right) \prod_{j=1}^{m-1} \left\{ \left(\prod_{\substack{k<0 \\ k \in \mathbb{Z}}} \frac{a+j\tau+k}{b+j\tau+k} \right) \left(\prod_{k \geq 0} \frac{a+j\tau+k}{b+j\tau+k} \right) \right\}$$

$$\cdot \left(\prod_{\substack{k<0 \\ k \in \mathbb{Z}}} \frac{a+m\tau+k}{b+m\tau+k} \right) \left(\prod_{k=0}^{n-1} \frac{a+m\tau+k}{b+m\tau+k} \right).$$

Regularizing the product

For the nonnegative part, write

$$\prod_{0\le k<N}\frac{a+k}{b+k}=\left(\prod_{0\le k<N}\frac{a+k}{b+k}e^{-\frac{a-b}{k+1}}\right)e^{(a-b)\zeta_{N+1}(1)}.$$

Regularizing by replacing $\zeta_{N+1}(1)$ with 0 gives $\dfrac{\Gamma(b)}{\Gamma(a)}e^{\gamma(b-a)}$. For the negative part, write

$$\prod_{-N<k<0}\frac{a+k}{b+k}=\left(\prod_{-N<k<0}\frac{a+k}{b+k}e^{-\frac{a-b}{k}}\right)e^{-(a-b)\zeta_{N}(1)}.$$

Replacing $\zeta_N(1)$ with $-T$ gives $\dfrac{\Gamma(1-b)}{\Gamma(1-a)}e^{(a-b)(\gamma+T)}$.

Regularizing the product

Applying these regularizations to all factors and simplifying gives

$$\prod_{0\preceq\mu\prec\lambda}^{*T}\frac{a+\mu}{b+\mu}:=\left(\prod_{j=0}^{m-1}\frac{\Gamma(b+j\tau)\Gamma(1-b-(j+1)\tau)}{\Gamma(a+j\tau)\Gamma(1-a-(j+1)\tau)}\right)e^{m(a-b)T}\frac{(a+m\tau)_n}{(b+m\tau)_n}.$$

Using this, define

$$\prod_{0\preceq\mu\prec\lambda}^{*T}\frac{(a+\mu)(b+\mu)}{(1+\mu)(c+\mu)}:=\left(\prod_{0\preceq\mu\prec\lambda}^{*T}\frac{a+\mu}{1+\mu}\right)\left(\prod_{0\preceq\mu\prec\lambda}^{*T}\frac{b+\mu}{c+\mu}\right).$$

Hypergeometric series of Eisenstein type

Define the hypergeometric series of Eisenstein type by

$$ {}_2E_1^T\left[\begin{matrix}a, b\\ c\end{matrix}; \tau; 1\right] := \sum_{\substack{0 \preceq \lambda \\ \lambda = m\tau + n}} e^{2mT} \prod_{0 \preceq \mu \prec \lambda}^{*T} \frac{(a+\mu)(b+\mu)}{(1+\mu)(c+\mu)}. $$

The factor e^{2mT} is a normalization chosen to give the main theorem a simple form. Gauss's classical hypergeometric summation is

$$ {}_2F_1\left[\begin{matrix}a, b\\ c\end{matrix}; 1\right] = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}. $$

Here and throughout, $q = e^{2\pi i \tau}$. Define an analogue of the gamma function by

$$ \Gamma_\tau(a) := \Gamma(a) \frac{(q;q)_\infty}{(e^{2\pi i a} q; q)_\infty}. $$

An Eisenstein analogue of Gauss's summation

We write ${}_2E_1^- := {}_2E_1^{-\pi i}$. We have the following analogue of Gauss's hypergeometric summation.

Theorem (M. 2026+)

Let $\Im\tau > 0$. *Then*

$$ {}_2E_1^T\left[\begin{matrix}a, b\\ c\end{matrix}; \tau; 1\right] = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}\,{}_2\phi_1\left[\begin{matrix}e^{2\pi i a}, e^{2\pi i b}\\ e^{2\pi i c}q\end{matrix}; q, e^{(T-\pi i)(1+a+b-c)}q\right]. $$

In particular,

$$ {}_2E_1^-\left[\begin{matrix}a, b\\ c\end{matrix}; \tau; 1\right] = \frac{\Gamma_\tau(c)\Gamma_\tau(c-a-b)}{\Gamma_\tau(c-a)\Gamma_\tau(c-b)}. $$

These identities follow from Dougall's ${}_2H_2$ summation and Heine's q-Gauss summation. We omit the proof in this talk.

Our main theorem gives an analogue of the Ohno–Zagier relation.

Theorem (M. 2026+)

Let $\Im\tau > 0$, $\alpha + \beta = u + v$, and $\alpha\beta = w$. As a formal power series in u, v, w,

$$\sum_{k,r,s>0} \left(\sum_{\boldsymbol{k} \in I_0(k,r,s)} G^T(\boldsymbol{k};\tau) \right) u^{k-r-s} v^{r-s} w^{s-1}$$

$$= \frac{1}{uv - w} \left(1 - {}_2E_1^T \left[\begin{matrix} \alpha - u, \beta - u \\ 1 - u \end{matrix}; \tau; 1 \right] \right).$$

For $T = -\pi i$, the Gauss-type summation gives a closed form in terms of the gamma analogue.

Theorem (M. 2026+)

Let $\Im\tau > 0$, $\alpha + \beta = u + v$, and $\alpha\beta = w$. As a formal power series in u, v, w,

$$\sum_{k,r,s>0} \left(\sum_{\boldsymbol{k} \in I_0(k,r,s)} G^-(\boldsymbol{k};\tau) \right) u^{k-r-s} v^{r-s} w^{s-1}$$

$$= \frac{1}{uv - w} \left(1 - \frac{\Gamma_\tau(1-u)\Gamma_\tau(1-v)}{\Gamma_\tau(1-\alpha)\Gamma_\tau(1-\beta)} \right).$$

The gamma analogue has the expansion

$$\Gamma_\tau(1-x) = \exp\left[\left(\gamma - 2\pi i \sum_{n>0} \frac{q^n}{1-q^n}\right)x + \sum_{n\geq 2} \frac{G^-(n;\tau)}{n} x^n\right].$$

Thus the Ohno–Zagier-type relation shows that the sum of G^--type multiple Eisenstein series of fixed weight, depth, and height,

$$\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^-(\boldsymbol{k};\tau),$$

is a polynomial in the depth-one values $G^-(2;\tau), G^-(3;\tau), \ldots$.

Proof sketch of the Ohno–Zagier-type relation

First, the definition of $G^T(\boldsymbol{k};\tau)$ gives

$$\begin{aligned}\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^T(\boldsymbol{k};\tau) &= \sum_{\boldsymbol{k}\in I_0(k,r,s)} \sum_{(\boldsymbol{k}_1,\boldsymbol{k}_2)=\boldsymbol{k}} \zeta^*(\boldsymbol{k}_1) h^T(\boldsymbol{k}_2;\tau) \\ &= \sum_{\boldsymbol{k}\in I_0(k,r,s)} \zeta(\boldsymbol{k}) \\ &\quad + \sum_{\substack{k_1,r_1,s_1\geq 0\\ k_2,r_2,s_2>0\\ k_1+k_2=k\\ r_1+r_2=r\\ s_1+s_2=s}} \left(\sum_{\boldsymbol{k}_1\in I(k_1,r_1,s_1)} \zeta^*(\boldsymbol{k}_1) \right) \left(\sum_{\boldsymbol{k}_2\in I_0(k_2,r_2,s_2)} h^T(\boldsymbol{k}_2;\tau) \right).\end{aligned}$$

Proof sketch of the Ohno–Zagier-type relation

Passing to generating functions, we obtain

$$\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^T(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1} = \sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} \zeta(\boldsymbol{k})\right) u^{k-r-s}v^{r-s}w^{s-1}$$
$$+\left(\sum_{k,r,s\geq 0}\left(\sum_{\boldsymbol{k}\in I(k,r,s)} \zeta^*(\boldsymbol{k})\right) u^{k-r-s}v^{r-s}w^{s}\right)$$
$$\cdot\left(\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} h^T(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1}\right).$$

Proof sketch of the Ohno–Zagier-type relation

Combining the Ohno–Zagier relation with the known evaluation of Hirose–Murahara–Saito,

$$\sum_{k,r,s\geq 0}\left(\sum_{\boldsymbol{k}\in I(k,r,s)}\zeta^{*}(\boldsymbol{k})\right)u^{k-r-s}v^{r-s}w^{s}=\frac{e^{-\gamma v}\Gamma(1-u)}{\Gamma(1+\alpha-u)\Gamma(1+\beta-u)},$$

we obtain

$$\begin{aligned}&\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)}G^{T}(\boldsymbol{k};\tau)\right)u^{k-r-s}v^{r-s}w^{s-1}\\&=\frac{1}{uv-w}\left(1-\frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-\alpha)\Gamma(1-\beta)}\right)\\&\quad+\frac{e^{-\gamma v}\Gamma(1-u)}{\Gamma(1+\alpha-u)\Gamma(1+\beta-u)}\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)}h^{T}(\boldsymbol{k};\tau)\right)u^{k-r-s}v^{r-s}w^{s-1}.\end{aligned}$$

Proof sketch of the Ohno–Zagier-type relation

Next, a similar argument using the definition of $h^T(\boldsymbol{k};\tau)$ gives

$$\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} h^T(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1}$$

$$= \sum_{j\geq 1}\sum_{0<l_1<\cdots<l_j}\left(\sum_{\substack{k,r>0\\ s\geq 0}}\left(\sum_{\boldsymbol{k}\in I(k,r,s)}\Psi^T_{\boldsymbol{k}}(l_1\tau)\right)u^{k-r-s}v^{r-s}w^{s}\right)$$

$$\cdots\left(\sum_{\substack{k,r>0\\ s\geq 0}}\left(\sum_{\boldsymbol{k}\in I(k,r,s)}\Psi^T_{\boldsymbol{k}}(l_{j-1}\tau)\right)u^{k-r-s}v^{r-s}w^{s}\right)$$

$$\cdot\left(\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)}\Psi^T_{\boldsymbol{k}}(l_j\tau)\right)u^{k-r-s}v^{r-s}w^{s-1}\right).$$

Proof sketch of the Ohno–Zagier-type relation

A similar argument gives

$$\sum_{\substack{k,r>0\\ s\geq 0}} \left(\sum_{\boldsymbol{k}\in I(k,r,s)} \Psi_{\boldsymbol{k}}^{T}(t) \right) u^{k-r-s} v^{r-s} w^{s}$$

$$= \left(\sum_{k,r,s\geq 0} \left(\sum_{\boldsymbol{k}\in I(k,r,s)} \zeta^{-T}(\boldsymbol{k}; -t-1) \right) (-u)^{k-r-s} (-v)^{r-s} w^{s} \right)$$

$$\cdot \left(\sum_{k,r,s\geq 0} \left(\sum_{\boldsymbol{k}\in I(k,r,s)} \zeta^{*}(\boldsymbol{k}; t) \right) u^{k-r-s} v^{r-s} w^{s} \right) - 1.$$

Each factor is a gamma product, as in Hirose–Murahara–Saito (details omitted). Also,

$$\sum_{k,r,s>0} \left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} \Psi_{\boldsymbol{k}}^{T}(t) \right) u^{k-r-s} v^{r-s} w^{s-1}$$

can likewise be evaluated using Dougall's ${}_2H_2$ summation, as in the approach to the Ohno–Zagier relation.

Proof sketch of the Ohno–Zagier-type relation

The resulting expressions are

$$f(n) := e^{T v} \frac{\sin \pi(n\tau + \alpha - u) \sin \pi(n\tau + \beta - u)}{\sin \pi n\tau \sin \pi(n\tau - u)},$$

$$g(n) := \frac{e^{(T+\gamma)v} \pi^2 \Gamma(1-v)}{\Gamma(1-\alpha+u)\Gamma(1-\beta+u)\Gamma(1-\alpha)\Gamma(1-\beta) \sin \pi n\tau \sin \pi(n\tau - u)}.$$

With this notation,

$$\begin{aligned}
&\sum_{k,r,s>0} \left(\sum_{\boldsymbol{k} \in I_0(k,r,s)} h^T(\boldsymbol{k}; \tau) \right) u^{k-r-s} v^{r-s} w^{s-1} \\
&= \sum_{j \geq 1} \sum_{0 < l_1 < \cdots < l_j} (f(l_1) - 1) \cdots (f(l_{j-1}) - 1) g(l_j) \\
&= \sum_{n>0} \left(\prod_{k=1}^{n-1} f(k) \right) g(n), \qquad l_j \mapsto n.
\end{aligned}$$

Proof sketch of the Ohno–Zagier-type relation

Substituting the explicit formulas for f, g and rewriting in terms of q-shifted factorials gives

$$\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} h^T(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1}$$
$$= \frac{e^{\gamma v}\Gamma(\alpha-u)\Gamma(\beta-u)\Gamma(1-v)}{\Gamma(1-\alpha)\Gamma(1-\beta)}\left({}_2\phi_1\left[\begin{matrix}e^{2\pi i(\alpha-u)}, e^{2\pi i(\beta-u)}\\ e^{-2\pi iu}q\end{matrix}; q, e^{(T-\pi i)v}q\right]-1\right).$$

Combining this with the preceding calculations, we obtain

$$\sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^T(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1}$$
$$= \frac{1}{uv-w}\left(1-\frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-\alpha)\Gamma(1-\beta)}{}_2\phi_1\left[\begin{matrix}e^{2\pi i(\alpha-u)}, e^{2\pi i(\beta-u)}\\ e^{-2\pi iu}q\end{matrix}; q, e^{(T-\pi i)v}q\right]\right).$$

Proof sketch of the Ohno–Zagier-type relation

Using the Gauss-type formulas

$$ {}_2E_1^T\left[\begin{matrix}a, b\\ c\end{matrix}; \tau; 1\right] = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)} {}_2\phi_1\left[\begin{matrix}e^{2\pi i a}, e^{2\pi i b}\\ e^{2\pi i c}q\end{matrix}; q, e^{(T-\pi i)(1+a+b-c)}q\right], $$

$$ {}_2E_1^-\left[\begin{matrix}a, b\\ c\end{matrix}; \tau; 1\right] = \frac{\Gamma_\tau(c)\Gamma_\tau(c-a-b)}{\Gamma_\tau(c-a)\Gamma_\tau(c-b)}, $$

we can rewrite the result as

$$ \sum_{k,r,s>0}\left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^-(\boldsymbol{k};\tau)\right) u^{k-r-s}v^{r-s}w^{s-1} = \frac{1}{uv-w}\left(1-\frac{\Gamma_\tau(1-u)\Gamma_\tau(1-v)}{\Gamma_\tau(1-\alpha)\Gamma_\tau(1-\beta)}\right). $$

Aoki–Ohno- and Aoki–Kombu–Ohno-type identities

Let $G^{-,\star}(\boldsymbol{k};\tau)$ be the sum of $G^{-}(\boldsymbol{I};\tau)$ over all indices $\boldsymbol{I}$ obtained by merging adjacent components of $\boldsymbol{k}$, and set

$$\Phi^{-,\star}(u,v,w;\tau) := \sum_{k,r,s>0} \left(\sum_{\boldsymbol{k}\in I_0(k,r,s)} G^{-,\star}(\boldsymbol{k};\tau) \right) u^{k-r-s} v^{r-s} w^{s-1}.$$

Theorem (Aoki–Kombu–Ohno type)

$$\Phi^{-,\star}(u,v,w;\tau) = \sum_{\lambda\succ 0} \frac{\lambda}{(\lambda-v)((\lambda-u)(\lambda-v)-w)} \prod_{\mu\succ\lambda} \frac{\mu(\mu-u-v)}{(\mu-u)(\mu-v)-w}.$$

Theorem (Aoki–Ohno type)

$$\Phi^{-,\star}(u,u,w;\tau) = 2 \sum_{m\tau+n\succ 0} \frac{(-1)^{m+n-1} e^{\pi i m^2\tau - 2\pi i m u}}{(m\tau+n-u)^2-w}.$$

Here $\lambda,\mu \in \mathbb{Z}+\mathbb{Z}\tau$ and $m,n\in\mathbb{Z}$; the identities are interpreted coefficientwise in the formal variables.